

CORE ELECTRONICS & TELECOMMUNICATION ENGINEERING

30-Page Foundation & Interview Revision Guide

Circuits • Devices • Digital • Signals • Communication • Control • Embedded • IoT

For B.Tech E&TC; students, placement preparation, viva revision and technical interviews.

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1. Electronics Fundamentals

Electrical quantities

Voltage, current, resistance, capacitance and inductance are the basic circuit quantities. Power is the rate of electrical energy transfer.

$$V = IR \quad P = VI = I^2R = V^2/R$$

Charge is measured in coulombs, current in amperes, resistance in ohms, capacitance in farads and inductance in henries.

DC and AC

DC maintains polarity in the ideal case. AC varies with time; a sinusoid is described by amplitude, frequency and phase.

$$v(t) = V_m \sin(\omega t + \phi), \quad \omega = 2\pi f$$

Laws

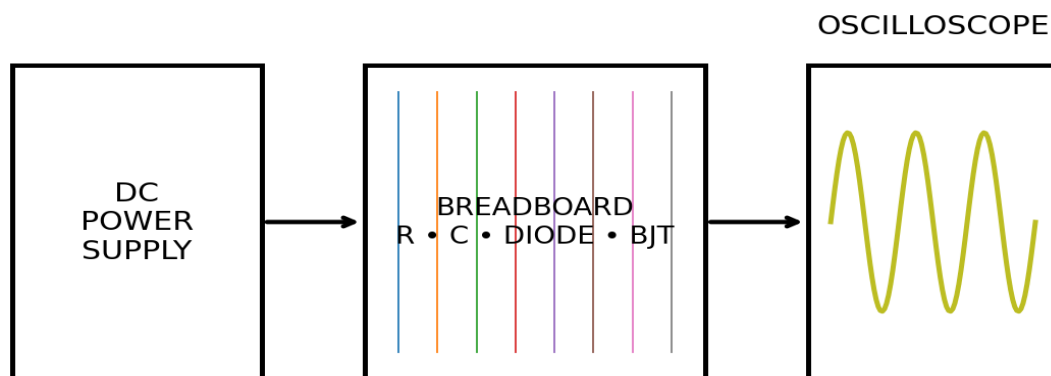
KCL: algebraic sum of currents at a node is zero. KVL: algebraic sum of voltages around a closed loop is zero.

$$\sum I = 0 \quad \sum V = 0$$

Quick points

• Open circuit → very high resistance. • Short circuit → very low resistance. • Fuse and circuit breakers protect against excessive current.

Typical Electronics Measurement Setup



Electronics laboratory measurement setup

2. Network Theorems & Analysis

Methods

Nodal analysis solves node voltages using KCL. Mesh analysis solves loop currents using KVL. Superposition applies to linear circuits.

Thevenin & Norton

Thevenin: V_{th} in series with R_{th} . Norton: I_n in parallel with R_n . They simplify complex two-terminal networks.

V_{th} = open-circuit voltage; R_{th} = resistance seen at the terminals.

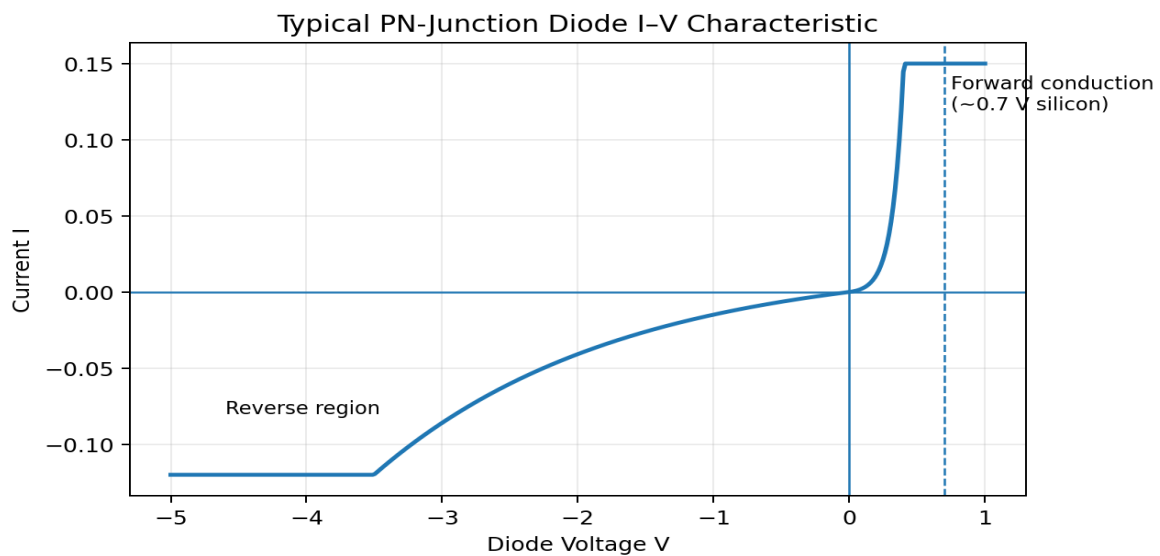
Maximum power

For a resistive load, maximum power transfer occurs when R_L equals R_{th} .

$$P_{max} = V_{th}^2 / (4R_{th})$$

Workflow

• Mark reference node. • Define polarities. • Write equations. • Solve. • Check units and limiting cases.



Typical diode current-voltage characteristic

3. Semiconductor Physics & Diodes

Semiconductors

Silicon is widely used. Doping produces n-type material with electrons as majority carriers and p-type material with holes as majority carriers.

PN junction

A depletion region forms at a p-n junction. Forward bias reduces the barrier; reverse bias widens it.

Diode types

• Rectifier diode: AC-to-DC conversion. • Zener: regulation/reference. • LED: light emission. • Photodiode: optical detection.

Equation

$$I = I_S [e^{(V/(nVT))} - 1]$$

The actual forward drop depends on current, temperature and device technology.

4. Rectifiers, Filters & Power Supplies

Rectifiers

Half-wave uses one half-cycle. Full-wave uses both. A bridge rectifier uses four diodes and avoids a center-tapped transformer.

Filters

Capacitor-input filters store charge and reduce ripple. LC filters can provide stronger smoothing.

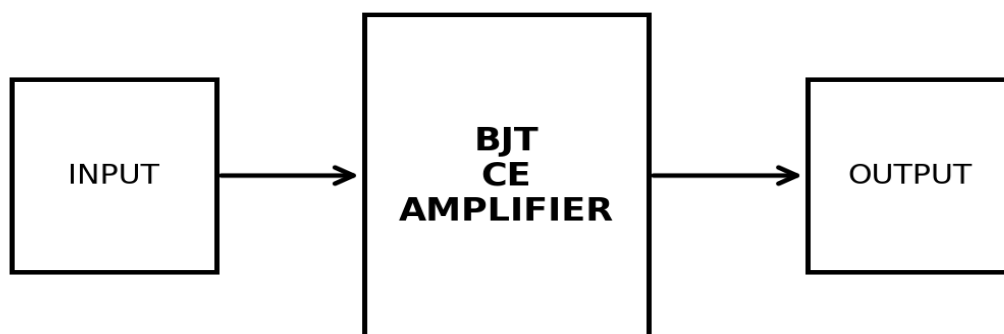
Regulation

Linear regulators are simple and low-noise but dissipate heat. Switching regulators are usually more efficient and are common in SMPS.

Design checks

• Input/output voltage • Load current • Ripple • Thermal dissipation • Transient response • Protection

Biasing + Small-Signal Amplification



Q-point keeps the transistor in the active region for linear amplification.

BJT common-emitter amplifier block

5. BJT Transistors

Operation

A BJT has emitter, base and collector. In active mode, base current controls collector current.

$$I_C \approx \beta I_B \quad I_E = I_C + I_B$$

Configurations

Common-emitter gives useful voltage/current gain. Common-collector acts as an emitter follower. Common-base has low input impedance and good high-frequency behavior.

Regions

• Cutoff: OFF. • Active: amplification. • Saturation: strongly ON and useful for switching.

Interview focus

Biasing, Q-point, load line, β , thermal stability and CE amplifier behavior are essential.

6. FETs, MOSFETs & Power Devices

FET/MOSFET

FETs are voltage-controlled devices. MOSFET gates are insulated, giving very high input impedance. Enhancement MOSFETs are normally OFF at zero gate bias.

Power MOSFET

Used in SMPS, motor drives, battery systems and converters. Important ratings: V_{DS} , I_D , $R_{DS(on)}$, gate charge and thermal resistance.

IGBT and thyristors

IGBTs suit higher-voltage/current switching. SCR, TRIAC and DIAC devices are common in controlled power electronics.

7. Operational Amplifiers

Ideal assumptions

Infinite open-loop gain and input impedance, zero output impedance and infinite bandwidth are idealizations. With negative feedback, $V_+ \approx V_-$ and input currents are approximately zero.

Key circuits

Inverting gain: $A_v = -R_f/R_{in}$

Non-inverting gain: $A_v = 1 + R_f/R_1$

Applications include summing amplifiers, integrators, differentiators, comparators, active filters and instrumentation amplifiers.

Non-ideal effects

Offset voltage, bias current, slew rate, gain-bandwidth product, common-mode range and output swing matter in real designs.

CMRR

High common-mode rejection is important when extracting small sensor signals in the presence of common interference.

8. Oscillators & Timers

Oscillator

An oscillator generates a periodic waveform without a periodic external input. Feedback must satisfy the required gain and phase conditions.

Barkhausen

$|A\beta| \approx 1$ and total phase shift $\approx 0^\circ$ or 360°

Types

RC: lower frequencies. LC: higher frequencies. Crystal: high stability. Relaxation: square/triangle-like waveforms.

555 timer

The 555 operates in monostable and astable modes for delays, pulse generation, blinking and timing.

9. Analog Modulation

Purpose

Modulation shifts a baseband signal to a suitable frequency range for radiation, antenna practicality, spectrum allocation and multiplexing.

AM/FM/PM

AM varies carrier amplitude. FM varies carrier frequency. PM varies carrier phase.

$$\text{AM: } s(t) = A_c[1 + \mu \cos(\omega_m t)] \cos(\omega_c t)$$

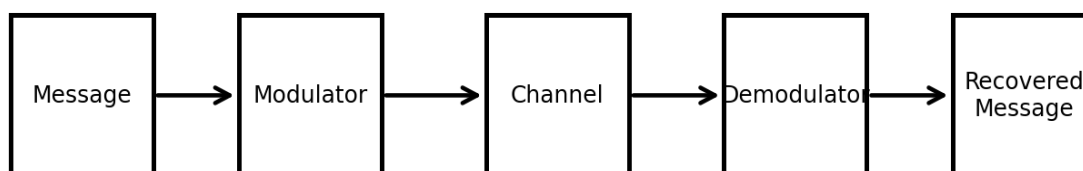
Comparison

AM is simple but amplitude-noise sensitive. FM generally has better noise performance but uses more bandwidth. PM is closely related to FM.

Terms

Know carrier frequency, modulation index, sidebands, bandwidth, frequency deviation, pre-emphasis and de-emphasis.

Basic Communication System



Noise and interference may enter through the channel.

Basic communication system

10. Digital Communication

Digital signals

Digital systems represent information using discrete symbols/bits, enabling regeneration, error detection/correction and reliable processing.

Sampling

Sampling converts continuous-time signals into samples. Quantization maps amplitudes to finite levels.

$f_s \geq 2f_m$ for a band-limited signal with highest frequency f_m

Digital modulation

ASK changes amplitude, FSK changes frequency, PSK changes phase, and QAM changes amplitude and phase together.

Performance

BER, SNR, bit rate, symbol rate, bandwidth and constellation spacing are key measures.

11. Pulse Modulation & PCM

PAM/PWM/PPM

PAM varies pulse amplitude, PWM varies pulse width and PPM varies pulse position.

PCM

Typical chain: anti-alias filter → sampling → quantization → encoding → transmission → decoding → reconstruction.

Quantization levels = 2^N

Delta modulation

A one-bit difference signal is transmitted. Slope overload and granular noise are key limitations. Adaptive delta modulation varies step size according to signal behavior.

12. Channels & Noise

Media

Guided: twisted pair, coaxial cable, optical fiber. Unguided: radio, microwave and satellite.

Noise

Thermal noise comes from random carrier motion. Shot noise comes from discrete charge flow. Interference can arise from switching, other transmitters and multipath.

SNR

$$\text{SNR} = P_s/P_n; \text{SNR(dB)} = 10 \log_{10}(P_s/P_n)$$

Link budgets consider transmit power, antenna gain, path loss, cable loss, receiver sensitivity and noise.

Fading

Multipath causes interference and delay spread. Diversity, equalization and robust modulation help reduce its impact.

13. Antennas & Propagation

Antenna parameters

Radiation pattern, gain, directivity, efficiency, polarization and bandwidth are key parameters.

Types

Dipole/monopole: basic radiators. Yagi-Uda: directional. Horn: microwave. Parabolic reflector: high-gain narrow beam.

Propagation

Ground wave, sky wave and line-of-sight propagation dominate different situations. Free-space path loss increases with distance and frequency.

14. EM Waves & Transmission Lines

EM waves

Maxwell's equations relate electric fields, magnetic fields, charge and current. EM waves can propagate through free space.

Transmission lines

At high frequencies, wires behave as distributed systems with characteristic impedance, delay and reflections.

$$\Gamma = (Z_L - Z_0) / (Z_L + Z_0)$$

VSWR

VSWR indicates mismatch severity. A perfectly matched line has VSWR = 1.

15. Microwave Engineering

Applications

Microwave systems support high bandwidth and directional links for radar, satellite communication, wireless backhaul and sensing.

Components

Waveguides guide microwave energy. Klystrons and magnetrons are vacuum-tube sources. Gunn and IMPATT devices are semiconductor microwave sources. Isolators and circulators control signal flow.

Design focus

Impedance matching, filters, oscillators, low-noise amplifiers, mixers and power amplifiers are important RF blocks.

16. Digital Electronics & Logic

Number systems

Binary, octal, decimal and hexadecimal are essential. Practice conversion and signed-number representation.

Gates

AND, OR and NOT are basic. NAND and NOR are universal. XOR is useful for addition/parity; XNOR is useful for equality detection.

Boolean laws

$$A+0=A; \quad A \cdot 1=A; \quad A+A'=1; \quad A \cdot A'=0$$

$$(AB)'=A'+B'; \quad (A+B)'=A'B'$$

17. Combinational Circuits

Blocks

Adders, subtractors, multiplexers, demultiplexers, encoders, decoders and comparators are combinational circuits.

Full adder

A full adder adds A, B and carry-in.

$$\text{Sum} = A \oplus B \oplus C_{in}$$

$$\text{Cout} = AB + BC_{in} + AC_{in}$$

MUX

A multiplexer selects one of several inputs using select lines and forwards it to an output.

18. Sequential Circuits & Memory

Flip-flops

SR, JK, D and T flip-flops store one bit. D is common in registers; T is useful for counters.

Registers/counters

Registers store and shift groups of bits. Counters progress through states and may be synchronous or asynchronous.

Memory

RAM is generally volatile. ROM variants retain data without power. Cache is smaller and faster than main memory.

Timing

Setup time, hold time, propagation delay, clock skew and metastability are core concepts in synchronous design.

19. Microprocessors & Microcontrollers

Difference

A microprocessor usually needs external memory/peripherals. A microcontroller integrates CPU, memory, timers, GPIO and communication interfaces.

8051

The classic 8051 family provides an 8-bit CPU, GPIO, timers/counters, serial communication and interrupts. Keil is a common development environment.

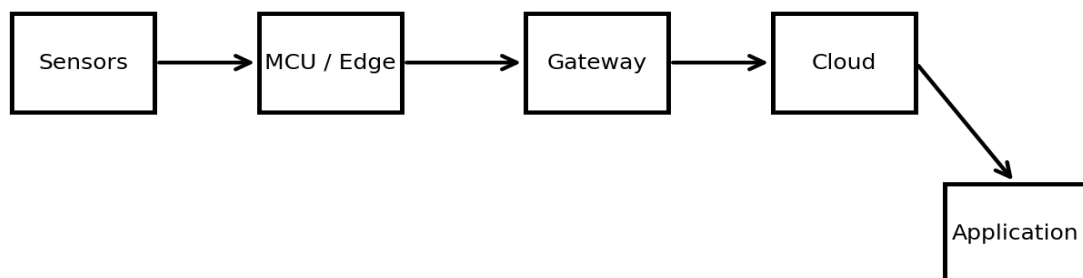
Modern MCUs

ARM Cortex-M controllers are widely used because of performance, low power and extensive peripherals.

Must know

Interrupts, timers, GPIO, ADC, PWM, watchdog, memory mapping and programming/debugging methods.

Embedded + IoT Data Flow



Sensing → Processing → Connectivity → Cloud → User

Embedded and IoT data flow

20. Embedded Systems

Definition

An embedded system is a dedicated computing system inside a larger product, combining hardware, firmware and interfaces.

Real time

Hard real-time systems have strict deadlines. Soft real-time systems tolerate occasional deadline misses.

Design flow

Requirements → architecture → hardware/firmware → integration → testing → validation. Consider cost, power, memory, latency, reliability and safety.

Interfaces

UART, SPI, I²C, CAN, USB, Ethernet and GPIO are widely used.

21. Sensors & Instrumentation

Sensor

A sensor converts a physical quantity into a measurable signal. Common quantities include temperature, pressure, level, light, displacement and acceleration.

Examples

RTD/thermistor: temperature. Strain gauge: strain. LDR/photodiode: light. Accelerometer: acceleration. Ultrasonic: distance. Pressure sensor: pressure.

Instrumentation chain

Physical variable → sensor → signal conditioning → ADC → processor → decision/control → display/communication.

22. ADC, DAC & Signal Conditioning

ADC

ADC converts analog input into digital code. Specifications include resolution, sampling rate, INL/DNL, SNR, reference voltage and latency.

Ideal quantization step $\approx VFS/2^N$

DAC

DAC converts a digital code to an analog level. R-2R ladder and current-steering architectures are common.

Conditioning

Amplification, attenuation, filtering, isolation, linearization and level shifting prepare sensor signals. Anti-alias filtering should precede sampling.

23. Control Systems

Open/closed loop

Open-loop systems do not use feedback. Closed-loop systems compare output with reference and act on error.

Transfer function

$$G(s) = \text{Output}(s) / \text{Input}(s)$$

Poles strongly influence stability and transient behavior; zeros shape response.

PID

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

P responds to present error, I reduces steady-state error, D anticipates change but can amplify noise.

24. DSP Fundamentals

DSP

Digital signal processing analyzes, transforms and filters sampled signals using algorithms.

Aliasing

Sampling too slowly causes spectral overlap. Anti-alias filtering and adequate sampling frequency prevent it.

Filters

Low-pass passes low frequencies; high-pass passes high; band-pass passes a band; band-stop rejects a band. FIR can provide exact linear phase; IIR can achieve sharp responses with fewer coefficients.

Transforms

The DFT gives frequency-domain information for finite sequences. FFT is an efficient algorithm for computing the DFT.

25. PLC, SCADA & Automation

PLC

A PLC is a rugged industrial controller. It reads inputs, executes logic and updates outputs cyclically.

SCADA

SCADA supervises industrial processes, collects data, shows trends/alarms and supports supervisory control.

Sensors/signals

Pressure, temperature, level, flow, proximity and photoelectric sensors are common. Industrial signals include digital I/O and 4–20 mA analog loops.

VFD

A variable-frequency drive controls AC motor speed mainly by controlling frequency and voltage, improving process control and often energy efficiency.

26. IoT & Industrial IoT

Architecture

Typical flow: sensing → connectivity → edge/gateway → cloud/platform → application. Security and device management span the architecture.

Protocols

MQTT: lightweight publish/subscribe. CoAP: lightweight request/response for constrained devices. HTTP/HTTPS: web communication. AMQP: message-oriented enterprise communication.

Edge/fog

Edge processing occurs close to the data source to reduce latency and bandwidth. Fog extends distributed processing between edge and centralized cloud systems.

27. Wireless & Mobile Communication

Cellular

Networks divide areas into cells and reuse spectrum. Handover maintains service while a device moves between cells.

Generations

2G brought digital cellular services. 3G improved mobile data. 4G LTE emphasized high-speed packet data. 5G adds enhanced mobile broadband, low-latency capabilities and massive machine communication.

Key concepts

Frequency reuse, OFDM, MIMO, beamforming, handover, throughput, latency and spectrum efficiency are important interview topics.

28. Fiber Optics

Structure

Optical fiber has a core, cladding and protective coating. Light is guided when the conditions for total internal reflection are satisfied.

Types

Single-mode fiber suits long-distance/high-bandwidth links. Multimode is common for shorter links and local networks.

Components

LEDs/laser diodes can be sources. PIN and avalanche photodiodes are receivers. Optical amplifiers extend link reach.

Advantages

High bandwidth, low electromagnetic interference, low attenuation and electrical isolation.

29. RF, Radar & Satellite

Radar

Radar transmits electromagnetic energy and analyzes echoes to estimate range, direction and sometimes velocity.

Range $\approx ct/2$ for round-trip delay t .

Satellite

A satellite link uses uplink, transponder and downlink. Frequency bands, antenna gain, link budget, propagation delay and rain attenuation affect performance.

RF front end

Important blocks include filters, low-noise amplifiers, mixers, oscillators, power amplifiers and matching networks.

30. Interview Revision & Career Checklist

Must-know formulas

$$V=IR$$

$$P=VI$$

$$f=1/T$$

$$\omega=2\pi f$$

$$A_v(\text{inverting})=-R_f/R_{in}$$

$$A_v(\text{non-inverting})=1+R_f/R_1$$

$$f_s \geq 2f_m$$

$$\Gamma=(Z_L-Z_0)/(Z_L+Z_0)$$

$$\text{Quantization levels}=2^N$$

Placement questions

• Explain your final-year project and exact contribution. • Compare BJT and MOSFET. • Why is negative feedback used? • Explain ADC and sampling theorem. • Microprocessor vs microcontroller? • Explain PLC, SCADA and VFD. • What is MQTT? • Compare AM, FM and digital modulation. • What is impedance matching? • Explain closed-loop control.

Revision strategy

Understand the concept first, memorize only the essential equations, then connect each topic to one practical application. For interviews, practice drawing simple block diagrams and explaining signal flow from input to output.